

ST Aqr: Photometric and Orbital Period Analysis of a Near-Contact Binary with a Secularly Decreasing Period

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Abstract

Near contact binary (NCB) systems are astrophysically quite interesting; the light curve (LC) of ST Aqr, a member of this class, was obtained from the Transiting Exoplanet Survey Satellite (TESS) database and analyzed in detail in this article. A q-search was undertaken to determine the mass ratio ($q = M_2/M_1$) for ST Aqr, and the q was found to be 0.472 ± 0.009 . It was showed that each body in the ST Aqr system is a NCB with a filling factor of approximately 99 percent. As accurate absolute parameters are essential for investigating the evolutionary properties of such systems, the body masses of ST Aqr were estimated as $M_1 = 1.81 \pm 0.10 M_\odot$ and $M_2 = 0.85 \pm 0.06 M_\odot$, while their corresponding radii were found to be $R_1 = 2.27 \pm 0.05 R_\odot$ and $R_2 = 1.67 \pm 0.04 R_\odot$. In the analysis of ST Aqr's orbital period (OP), a decrease in the period was detected, and this value was calculated to be 2.08 seconds per century. Since the bodies do not fill their Roche lobes, we believe that the decrease in OP cannot be explained by a conserved mass transfer between the bodies. For this reason, the observed orbital period decrease may be attributed to angular momentum loss (AML) from the system through magnetic braking associated with magnetic activity. The non-conservative mass transfer rate was calculated to be $1.64 \times 10^{-7} M_\odot$ per year. To better understand the nature of ST Aqr, it was plotted on the $\log M_{\text{tot}} - \log J$ and Hertzsprung-Russell (HR) diagrams.

Keywords: binaries, eclipsing - stars, near-contact binaries - stars,

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1. Introduction

Near contact binaries (NCBs) were defined by Shaw (1990) as a class of close binaries. Shaw (1994) reported that NCBs, which have orbital periods (OPs) shorter than one day (though there are rare exceptions that slightly exceed this limit), exhibit strong tidal interactions between their bodies. Although their facing surfaces are close, they are binary stars that are not in physical contact, such as W UMa-type systems (Shaw, 1994). In these systems, one component is typically a hot star of spectral type A or F, while the cool component is 1-2 spectral classes later (cooler) than the hot component (Shaw, 1994). In such systems, at least one of the bodies has filled or is about to fill its Roche lobe. Regarding NCBs, it has been proposed that they are the progenitors of A-type W UMa stars; Shaw et al. (1996) cited as reasons that NCBs are in the early stages of mass transfer and that their X-ray luminosity functions are similar to those of A-type W UMa stars. Since these systems are not in contact and therefore have not reached thermal equilibrium, the temperature difference is greater than in contact binaries. Shaw et al. (1996) examined 58 NCB systems using ROSAT data and found that their X-ray luminosity functions were similar to those of A-type W UMa systems, thereby supporting the scenario that NCBs could be the evolutionary precursors of A-type W UMa systems. Furthermore, Shaw et al. (1996) noted that the 14 identified NCB examples suggest that the single, apparent A-F-type stars observed in RASS may actually be binary systems with cold, X-ray-emitting companions, and proposed photometric methods for the discovery of new NCB candidates. Since determining the physical parameters of NCB candidate systems and reexamining them with precise data would provide valuable insights into our understanding of such systems, the ST Aqr system was selected and subjected to a detailed photometric analysis.

ST Aqr (TIC 38378877, TYC 5232-397-1, HD 211965, Gaia DR3 2622626156150704896), was discovered by Leavitt at Harvard in 1908 (Pickering, 1908) and classified as

an Algol-type variable star. Zacharov (1930) determined its OP to be 0.7810161
 30 days; based on the appearance of its LC, ST Aqr was classified as β Lyr. The
 spectral type of ST Aqr was proposed by Roman (1956) as A7+G8 IV. The light
 variations of ST Aqr were studied photoelectrically in yellow light by Knipe
 (1971), and later, light variations were obtained photoelectrically using UBV
 filters by Gleim (1973). The light curves (LCs) obtained by Gleim (1973) were
 35 analyzed by Mardirossian et al. (1980) using the Wood method and interpreted
 to indicate that both bodies of ST Aqr are in contact. Al-Naimiy and Fleyeh
 (1989) observed ST Aqr in BV filters and analyzed the LCs using a theoretical
 approach based on Fourier transforms. Finally, Jassur et al. (2000) analyzed
 LCs obtained in UBV filters and determined that ST Aqr's cold star fills its
 40 Roche lobe, while the hot star is in the process of doing so. Furthermore, Jas-
 sur et al. (2000) reported that both bodies are evolved stars that have passed the
 main-sequence phase. There are no published studies in the literature regarding
 changes in the OP of ST Aqr, and the most recent LC analysis was conducted
 in 2000. For this reason, the ST Aqr system was selected; detailed photometric
 45 analyses were performed, fundamental astrophysical parameters were estimated,
 and an effort was made to understand the nature of the ST Aqr.

2. Data Information

High-sensitivity photometric data obtained by NASA's TESS mission were
 used to investigate the photometric properties and OP variations of the ST
 50 Aqr system. The TESS (Ricker et al., 2015) observations were archived by
 MAST¹. TESS observations of the ST Aqr were obtained from two different
 sectors (sectors 42 and 92), and LCs were acquired in the visual region with
 exposure times of 600 and 200 seconds, respectively, in both sectors. TESS ob-
 servations were conducted in August 2021 and May 2025, with LC observations
 55 spanning approximately one month each. Because TESS observations provide

¹MAST, <https://archive.stsci.edu>

high-resolution photometric measurements, they serve as an important data source for identifying short-term variations and eclipse minima. The LCs from the two different sectors used for ST Aqr were first checked for data quality; measurements with low reliability were excluded from the analysis, and only
60 photometric data points suitable for analysis were used. In this context, the eclipse minima of the system were determined from the LCs obtained from the two different sectors, and the observation times corresponding to these minima were calculated (see Table 1). Thus, in addition to the minima times (MTs) previously published in the literature (O-C Gateway (Paschke and Brat,
65 2006)), the new MTs derived from TESS observations were also included in the study's dataset. To determine the MTs derived from the observational data, a polynomial fit was conducted using the least-squares technique. A polynomial function was fitted to the data points surrounding the minimum of the observed LC, and the minimum time value was obtained by identifying the point where
70 the derivative of the polynomial equals zero. This method allowed for a more precise determination of the minimum point on the LC, independent of the observation sampling interval. The obtained MTs are listed in Table 1 along with their errors. TESS data were used for two purposes: (i) to analyze the high-precision photometric LC of ST Aqr and determine the system's current
75 photometric properties, and (ii) to derive new MTs from the TESS LCs, combine these times with the existing MTs, and thereby investigate the system's OP variation over a longer time span. This approach allows for a combined assessment of both the current photometric structure and the long-term orbital evolution of ST Aqr. No radial velocity data are available for the ST Aqr.

80 3. Light Curve Analysis

LC analysis is one of the fundamental astrophysical analysis methods used to determine the physical properties of binaries by examining the time-dependent variations in their brightness. Information about the system can be obtained by analyzing the shape of the LC. In eclipsing binary systems in particular,

Table 1: MTs and errors calculated for ST Aqr from TESS observations.

$\text{BJD}_{\text{TDB}} (2400000+)$	Error	Type (I/II)	$\text{BJD}_{\text{TDB}} (2400000+)$	Error	Type (I/II)
59447.8684	0.0223	II	60820.0592	0.0447	II
59448.6502	0.0219	II	60820.8402	0.0625	II
59449.4305	0.0367	II	60821.6208	0.0375	II
59450.2116	0.0154	II	60822.4017	0.0557	II
59450.9923	0.0119	II	60822.9671	0.0215	II
59461.9263	0.0139	II	60823.9636	0.0519	II
59462.7067	0.0202	II	60824.7448	0.0439	II
59463.4885	0.0148	II	60825.5255	0.0561	II
59464.269	0.0342	II	60826.3065	0.0573	II
59465.0512	0.0203	II	60827.0874	0.0425	II
59465.8312	0.0199	II	60827.8685	0.0643	II
59467.3932	0.0175	II	60828.6497	0.0616	II
59468.1735	0.0273	II	60804.0467	0.0102	I
59468.9545	0.0759	II	60804.8276	0.0093	I
59469.7362	0.0193	II	60805.6087	0.0091	I
59449.0366	0.0376	I	60806.3897	0.0108	I
59450.5987	0.0207	I	60807.1707	0.0073	I
59451.3799	0.022	I	60807.9516	0.0097	I
59463.0944	0.0232	I	60808.7326	0.0101	I
59463.8754	0.0506	I	60810.2946	0.0124	I
59467.7803	0.0344	I	60811.0757	0.0078	I
59468.5615	0.0179	I	60811.8567	0.0077	I
59469.3422	0.0749	I	60812.6376	0.0087	I
59470.1233	0.0256	I	60813.4184	0.0056	I
60803.658	0.0568	II	60814.1993	0.0113	I
60804.4392	0.0391	II	60814.9804	0.0212	I
60805.22	0.0493	II	60818.1042	0.0081	I
60806.001	0.0558	II	60818.8852	0.008	I
60806.7819	0.048	II	60819.6662	0.0147	I
60807.5632	0.0636	II	60820.4472	0.009	I
60808.344	0.0766	II	60821.2282	0.0104	I
60809.1247	0.0465	II	60822.0092	0.0065	I
60809.9058	0.0467	II	60822.7903	0.0163	I
60810.6866	0.0485	II	60823.5712	0.0093	I
60811.4676	0.0492	II	60824.3522	0.0073	I
60812.2489	0.0427	II	60825.1333	0.0082	I
60813.0298	0.0449	II	60825.9143	0.0099	I
60813.8109	0.0529	II	60826.6953	0.0098	I
60814.5921	0.0389	II	60827.4763	0.0069	I
60818.4969	0.0697	II	60828.2571	0.008	I
60819.2782	0.0625	II			

LC analysis allows for the determination of the stars' orbital inclination, radius ratios, temperature ratios, luminosity contributions, and geometric structures. In current studies, LC analysis is performed using the Wilson-Devinney (WD) model and similar techniques. Thus, observational data are compared with theoretical models to calculate the most appropriate physical parameters of the system. In this study, the WD (Wilson and Devinney, 1971; van Hamme and Wilson, 2003) technique was used for LC analysis of data obtained from TESS. In the LC analysis performed using the WD technique, MOD 2 was selected for ST Aqr in accordance with the system's geometric structure. The results of the photometric solution particularly the initial parameters such as the effective temperature of the primary body (T_1) and the system's q are crucial for arriving at a solution. The T_1 was determined by considering the system's spectral type, and the value ($T_1 = 7745$ K) corresponding to spectral type A7 was taken from the tables in the study by Eker et al. (2020). This value was used as a fixed parameter in the LC analysis, while the temperature of the second body (T_2) was left as a free parameter during the solution and determined to provide the best fit between the observational LC and the theoretical LC. To determine the system's q , radial velocity analysis is preferred when radial velocity data are available; however, when a reliable spectral q is not available for the system, the photometric q -search method can be employed. In this context, a q -search was performed, and the q corresponding to the point where the value of $\sum(O - C)^2$ is minimal was accepted as the system's photometric q . During the photometric solution, the atmospheric properties of the stars were taken into account; for components with convective atmospheres, the gravitational darkening coefficient ($g = 0.32$) proposed by Lucy (1967) and Rucinski (1969) were used for bodies with convective atmospheres, while values of ($g = 1.0$) and ($A = 1.0$) were used for bodies with radiative atmospheres. The nonlinear edge darkening coefficients were taken from the tables provided by van Hamme (1993). During the solution, appropriate parameters were left free to obtain the theoretical LC that best represents the observational LC (see Figure 1), thereby determining the system's fundamental geometric and physical parameters. In

addition, the best theoretical model was obtained using cold-spot modeling of the secondary body. The parameters of the solution obtained by comparing the observational and theoretical LCs are listed in Table 2.

4. Orbital Period Analysis

OP analysis is a method that examines changes in the OP of multiple systems over time. The purpose of OP analysis is to identify the physical processes that influence the dynamic evolution of the systems and their OPs. The analysis is typically conducted using MTs. The differences between the observed MTs and the calculated MTs are plotted on an Observed-Calculated (O-C) diagram. Parabolic variations in the O-C diagram may indicate mass transfer or AML in the system, while periodic oscillations may suggest the presence of a third body or be caused by magnetic activity. This method is widely used in the study of the long-term dynamic behavior of eclipsing multiple systems. When high-precision minimum time data spanning multiple observation periods are used, even very small changes in the OP can be reliably determined. The MTs for ST Aqr, selected for this study, were prepared for analysis using data from the literature (O-C Gateway (Paschke and Brat, 2006)) and those determined in this study. The initial minimum time (T_0) and orbital period (P) values were obtained from the O-C Gateway (Paschke and Brat, 2006) and Atlas O-C (Kreiner, 2004). A decreasing parabolic variation was observed in the ST Aqr system and is expressed by Equation 1.

$$\text{Min I (BJD}_{\text{TDB}}) = T_0 + P \times E + Q \times E^2 \quad (1)$$

In Equation 1, E represents the number of epochs, and Q is the quadratic term. The decreasing parabolic trend is shown in Figure 2. This OP decrease rate, calculated as $dP/dt = 2Q/P$, was determined to be 2.41×10^{-7} days per year. Such changes may be attributed to either mass transfer or AML. Conservative and non-conservative mass transfer values have been calculated and are presented in Table 3. Since neither component of the ST Aqr system

Table 2: The LC was analyzed using the WD technique; below are some parameters and their associated errors for ST Aqr, as calculated from the analysis.

Parameters	Jassur et al. (2000)	This Study
i ($^{\circ}$)	73.15	75.59 (35)
T_1 (K)	8160.6	7745
T_2 (K)	5152.6	5274 (53)
Ω_1	2.6864	2.85 (4)
Ω_2	2.6385	2.82 (5)
q	0.3807	0.472 (9)
A_1	1	1
A_2	0.5	0.5
g_1	1	1
g_2	0.32	0.32
$L_1/(L_1 + L_2)$	0.9336 (149)	0.914 (7)
$L_2/(L_1 + L_2)$	0.0664	0.086
r_1 (mean)	0.4716	0.4592 (6)
r_2 (mean)	0.3174	0.3373 (8)
$FR1^*$ (%)	99	99
$FR2^*$ (%)	99	99
Spot Parameter	Secondary Body	
q (deg)	-	88 (5)
f (deg)	-	243 (7)
r (deg)	-	17 (2)
$T_f = T_d/T_s$	-	0.89
$\Sigma W(O - C)^2$	-	0.105

*FR1 and *FR2: Roche lobe filling ratios for the first and second bodies.

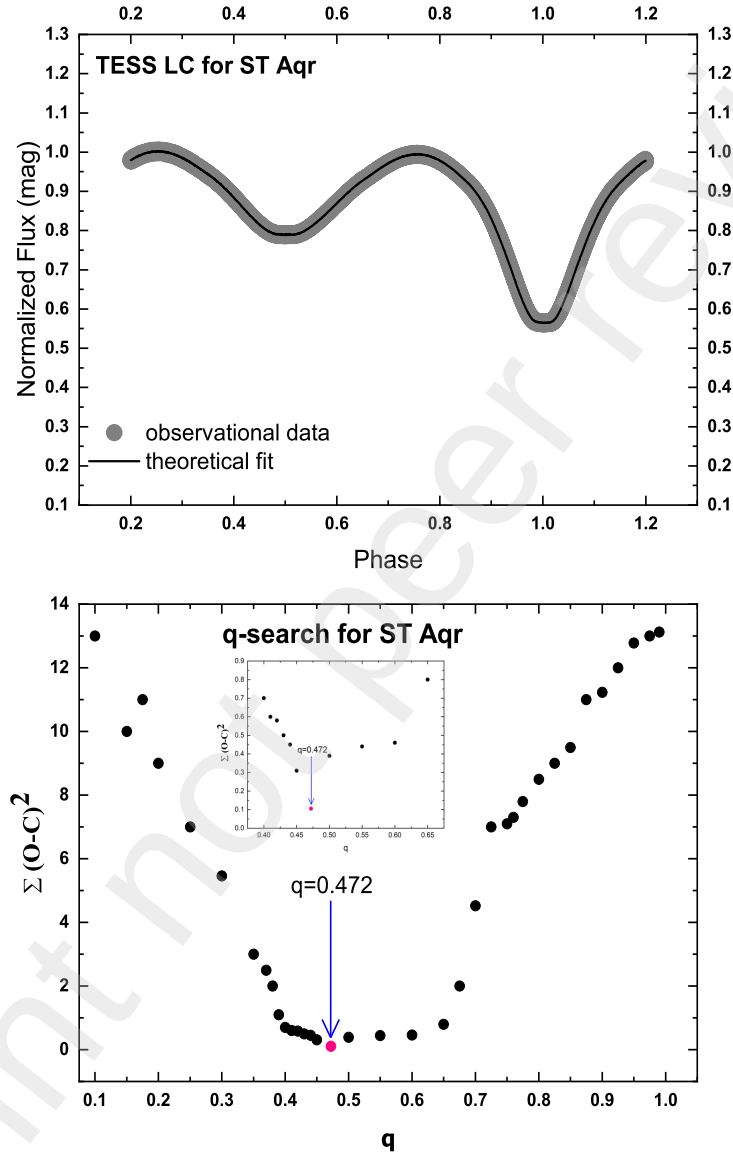


Figure 1: The agreement between the TESS LC for ST Aqr and the theoretical curve is shown in the top panel. The q-search for ST Aqr is shown in the bottom panel.

fills the Roche lobes (see Table 2 and Figure 3), non-conservative mass transfer may be proposed to explain this situation.

Table 3: Parameter values calculated from the O-C analysis of ST Aqr and their associated uncertainties.

Parameters	ST Aqr
T_0 (BJD _{TDB} +2400000)	41236.3121 (101)
P_{orb} (day)	0.7809976 (5)
Q (day)	$-2.58(4) \times 10^{-10}$
dP/dt (day/year)	2.41×10^{-7}
dM/dt ($M_{\odot}$ /year) (conservative)	1.65×10^{-7}
dM/dt ($M_{\odot}$ /year) (non-conservative)	1.64×10^{-7}

145 5. Results and Discussion

The reliable determination of photometric solutions for NCBs will enable a better understanding of the physical properties of these systems and enhance our knowledge of the evolutionary processes of binary stars in general. For this reason, the precise determination of photometric parameters will be crucial both
150 for a better understanding of binary star evolution and for understanding the statistical evolution of NCBs. In this context, the NCB ST Aqr was selected for this study, and its LC was analyzed in detail. No contribution from a third light (L_3) was detected in the LC analysis; instead, the solution was obtained by modeling a cold spot on the surface of the secondary object (see Figure 1). The
155 Roche geometry at the 0.75 phase is shown in Figure 3, and the cold spot on the second body is visible in red. Table 1 presents the parameters determined in this study alongside those determined by Jassur et al. (2000), whose solution did not include a cold spot model. A q-search was conducted to determine the q of ST Aqr, yielding a value of $q = 0.472$ (9) (see Figure 1), which differs from the
160 value determined by Jassur et al. (2000). Only a 2471 K difference was found between the surface temperatures of the objects, and it was determined that the

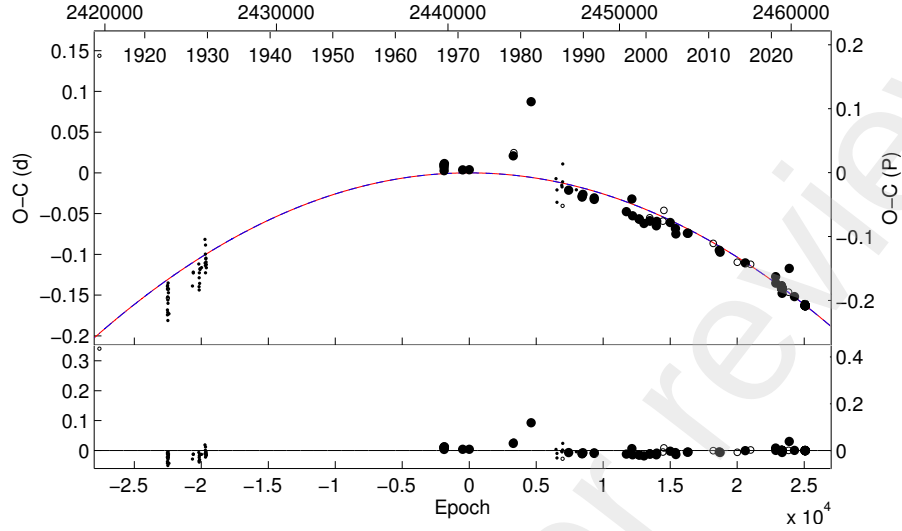


Figure 2: The O-C diagram for ST Aqr. A parabolic trend is observed, indicating a steady decrease in OP.

temperatures of the two bodies were quite far apart. The absence of any radial velocity studies on ST Aqr in the current literature makes the determination of ST Aqr's absolute parameters particularly important. Therefore, ST Aqr's absolute physical parameters and their associated error values were calculated using the relevant astrophysical equations, and the results are presented in Table 4. In determining the absolute parameters of ST Aqr, the values for the Sun's effective temperature, bolometric absolute magnitude, and surface gravity provided by Pecaut and Mamajek (2013) $T_{\text{eff}\odot} = 5771.8$ (7) K, $M_{\text{bol}\odot} = 4.7554$ (4) mag and $g_{\odot} = 27423.2$ (7.9) cm.s^{-2} were used. The bolometric correction (BC) values were taken from the study by Eker et al. (2020) and are presented in Table 4. Assuming that the primary body in ST Aqr is a main-sequence star, its mass is expected to be approximately $1.81 M_{\odot}$ based on the values provided by Eker et al. (2020). The error value for the mass of the primary body was evaluated as 10% of the mass. The absolute physical properties were derived from the appropriate astrophysical relations are listed in Table 4. A review of Table 4 reveals that these values differ from those estimated by Al-Naimiy

and Fleyeh (1989). The color index calculated using the method described by Schlafly and Finkbeiner (2011) is given in Table 4; thus, the distance to ST Aqr
180 was derived using the distance modulus and interstellar galactic absorption in
the form $A_v = 3.1E(B - V)$ (see Table 4). Finally, the distance value obtained
from the photometric analyses (see Table 4) was compared with the distance
provided in the Gaia DR3 (Gaia Collaboration, 2022) catalog, and the two val-
ues were found to be consistent with each other. This consistency supports the
185 reliability of the photometric solution obtained for ST Aqr and, consequently,
the parameters calculated from it.

Table 4: Absolute parameters for ST Aqr and the measurement uncertainties associated with these quantities.

Parameters	Al-Naimiy and Fleyeh (1989)	This Study
a ($R_\odot$)	4.57	4.95 (10)
M_1 ($M_\odot$)	1.50	1.81 (10)
M_2 ($M_\odot$)	0.60	0.85 (6)
R_1 ($R_\odot$)	2.24	2.27 (5)
R_2 ($R_\odot$)	1.19	1.67 (4)
$\log g_1$ (cgs)	—	3.98 (1)
$\log g_2$ (cgs)	—	3.92 (3)
$M_{\text{bol},1}$ (mag)	1.55	1.70 (16)
$M_{\text{bol},2}$ (mag)	4.87	4.26 (22)
L_1 ($L_\odot$)	19.36	16.42 (15)
L_2 ($L_\odot$)	0.91	1.54 (21)
BC_1	—	0.043
BC_2	—	-0.232
$E(B - V)$	—	0.074 (5)
$d_{\text{photometric}}$ (pc)	—	310 (36)
$d_{\text{Gaia-DR3}}$ (pc)	—	258 (1)

A total of 81 MTs were obtained from TESS observations of ST Aqr, including 37 for the first minima and 44 for the second minima. Additionally, 101 MTs

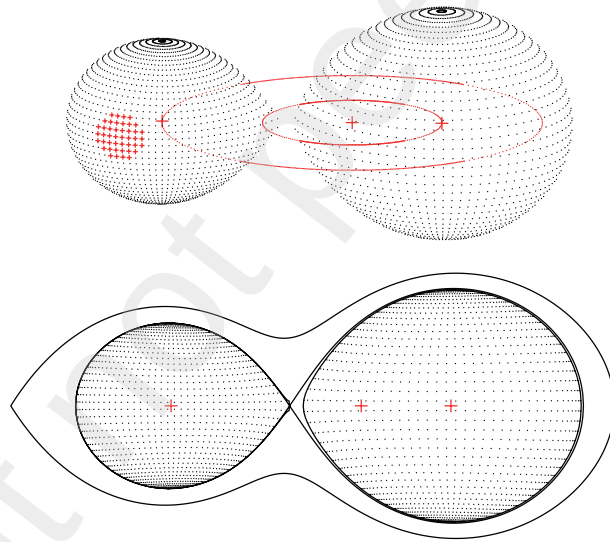


Figure 3: Roche geometry at phase 0.75, according to the photometric solution for ST Aqr.

(90 first minima and 11 second minima) were collected from the literature, and
 190 an OP analysis was performed using this data which, combined with the MTs
 obtained in this study, spans a broad time range of approximately 100 years.
 The OP analysis, which had not been previously conducted for ST Aqr, yielded
 a dP/dt value of 2.41×10^{-7} day/year. It was observed that the period of ST
 Aqr exhibits a long-term decrease (see Figure 2). The effect of mass loss and
 195 mass transfer between bodies on OP has been reported and applied by various
 authors (e.g., Tout and Hall (1991); Erdem and Öztürk (2014); Yang et al.
 (2016); Yıldırım and Soyduğan (2023)).

$$\frac{\dot{P}}{P} = \left[2 \left(\frac{R_a}{a} \right)^2 \frac{M_1 + M_2}{M_1 M_2} - \frac{2}{M_1 + M_2} \right] \dot{M} + \frac{3(M_1 - M_2)}{M_1 M_2} \dot{M}_1 \quad (2)$$

In Equation 2, $(\frac{\dot{P}}{P})$ represents the change in orbital period, (R_a) represents
 the Alfvén radius, and (a) represents the distance between the bodies. Tout and
 200 Hall (1991) assumed the Alfvén radius to be approximately 10 times the radius
 of the mass-losing system. M_1 and M_2 denote the masses of the bodies. The
 first term of the equation represents the effect of mass loss from the system on
 the OP, while the second term represents the effect of mass transfer between the
 bodies on the OP. Non-conservative mass transfer for ST Aqr was calculated
 205 and is listed in Table 3. Since neither of the two bodies in ST Aqr can fill their
 Roche lobes, the observed OP variation cannot be explained by mass transfer
 within the system. The spectral type of ST Aqr was proposed by Roman (1956)
 as A7+G8 IV; the secondary object is of the back-spectral type and is likely to
 exhibit magnetic activity. Additionally, a cold spot model was applied to the
 210 secondary body in the LC analysis conducted in this study. Furthermore, the
 log L_x value for ST Aqr, expressed in erg/s, was reported by Shaw et al. (1996).
 Taking all these factors into account, the observed OP decrease of approximately
 0.02 seconds per year in ST Aqr may be attributed to AML through magnetic
 braking associated with magnetic activity. For systems in which the second
 215 body fills the Roche lobe, or in which both bodies nearly fill the Roche lobe,
 the decrease in OP cannot be explained by an exchange of mass between the

bodies since the second bodies in NCBs are generally of the G or K spectral types; this decrease in OP may be due to AML caused by magnetic braking, as emphasized by Qian (2002). Yang et al. (2016) noted that the decrease in
 220 IZ Mon's OP is due to mass and AML, while Liao et al. (2017) noted that the long-term parabolic decrease in DI Hya's OP is due to AML via an enhanced stellar wind from the evolved secondary component. The IZ Mon and DI Hya systems provide supporting evidence for the explanation that the OP decrease of ST Aqr is due to AML.

225 Some NCBs like the ST Aqr type were selected from studies in the literature and listed in Table 5. The systems in Table 5 were selected because their analyses have been conducted in detail and their parameters have been determined with high precision. When examining the relationship between $\log M_{\text{tot}}$ and $\log J$, it is evident as also demonstrated by Eker et al. (2006) that there
 230 is a distinct boundary separating contact systems from discrete systems. The boundary line in Figure 4a is taken from Eker et al. (2006) and is given by the equation $\log J_{\text{lim}} = 0.522(\log M)^2 + 1.664(\log M) + 51.315$. The NCBs and ST Aqr shown in Figure 4a are seen to be located near this boundary region defined by Eker et al. (2006), and some systems are situated in the region of contact
 235 binaries. NCBs of A and F types with decreasing orbital periods (the ST Aqr system is also of spectral type A7 and has a decreasing OP), Qian (2002) reported that there may be a transition toward A and F type overcontact binaries with increasing OPs, and that NCBs could be precursors to A-type W UMa systems, oscillating around a weak contact state as predicted by the thermal
 240 relaxation oscillation (TRO) theory.

◆ The positions of ST Aqr and the NCBs in Table 5 on the HR diagram are shown in Figure 4b. The positions of the systems on the diagram were determined by taking into account the Padova evolutionary models developed by Bressan et al. (2012). The metal abundance value of $Z = 0.014$, which provides
 245 the best fit for the positions of ST Aqr and the NCBs on the HR diagram, was used to determine these positions. An examination of Figure 4b reveals that the primary bodies of these systems generally lie on the main sequence, while the

Table 5: Some astrophysical parameters of some NCBs compiled from the literature.

System	$M_1 (M_\odot)$	$M_2 (M_\odot)$	$R_1 (R_\odot)$	$R_2 (R_\odot)$	$T_1 (K)$	$T_2 (K)$	$L_1 (L_\odot)$	$L_2 (L_\odot)$	References
V1010 Oph	1.887	0.887	2.010	1.400	7500	5132	11.45	1.220	Siwak et al. (2010)
WZ Cyg	1.589	1.003	1.700	1.370	6530	4932	4.710	1.000	Siwak et al. (2010)
DO Cas	2.130	0.650	2.220	1.290	8350	4297	21.47	0.510	Siwak et al. (2010)
FS Lup	1.301	0.611	1.230	0.850	5860	5130	1.600	0.450	Siwak et al. (2010)
V747 Cen	1.541	0.493	1.700	0.980	8150	4275	11.42	0.290	Siwak et al. (2010)
II Per	0.950	0.360	1.310	0.840	5740	4464	6.860	0.250	Zhu et al. (2009)
V473 Cas	1.000	0.480	1.190	0.830	5830	4418	5.700	0.220	Zhu et al. (2009)
BS Vul	1.520	0.520	1.540	0.930	7000	4632	5.130	0.350	Zhu et al. (2012)
AX Dra	1.460	0.920	1.520	1.370	6850	4951	4.440	1.010	Kim et al. (2004)
ZZ Aur	1.620	0.760	1.730	1.260	7800	4978	9.780	0.860	Oh et al. (2006)
AS Ser	1.500	0.520	1.520	0.930	7000	4642	4.970	0.340	Zhu et al. (2008)
NP Aqr	1.650	0.990	2.130	1.670	7050	4250	10.00	0.832	Ibanoglu et al. (2010)
IR Cas	1.430	1.220	1.770	1.510	6750	5992	5.850	2.630	Li et al. (2014)
DI Hya	1.600	0.670	1.610	1.195	7300	4548	6.669	0.551	Liao et al. (2017)
IZ Mon	2.010	0.780	2.480	1.490	8500	4971	28.70	1.220	Yang et al. (2016)
HT Lyn	1.758	0.627	1.804	1.122	7768	5296	10.69	0.893	Meng et al. (2022)
IR Lyn	1.373	0.364	1.614	0.622	6720	4216	4.793	0.110	Meng et al. (2022)

secondary bodies lie off the main sequence. It is likely that the low-mass bodies are more evolved stars that have exhausted the hydrogen in their cores (Meng et al., 2022). Due to the marginal contact configuration, it is thought that NCBs may be in the phase predicted by TRO theory (Flannery (1976); Lucy (1976); Robertson and Eggleton (1977)). Considering the TRO theory, close binary stars must undergo TRO cycles near the marginal contact state in order to reach a state of overcontact and thermal equilibrium (Meng et al., 2022). In the approximately 100-year data set for ST Aqr, a continuous decrease in OP is observed, and since both bodies do not fill their Roche lobes but are very close to doing so (see Table 2), ST Aqr may serve as a unique test case for the TRO theory when evaluated from this perspective. Furthermore, this finding supports the idea that NCBs and Type A W UMa stars share similarities and that a transition may exist between them.

This study, conducted using TESS photometry of ST Aqr, presents the first comprehensive analysis of OP variations in the literature and demonstrates, with precise parameters, that the system is in a near-contact state. The calculated current mass, radius, and orbital parameters provide a contribution to the

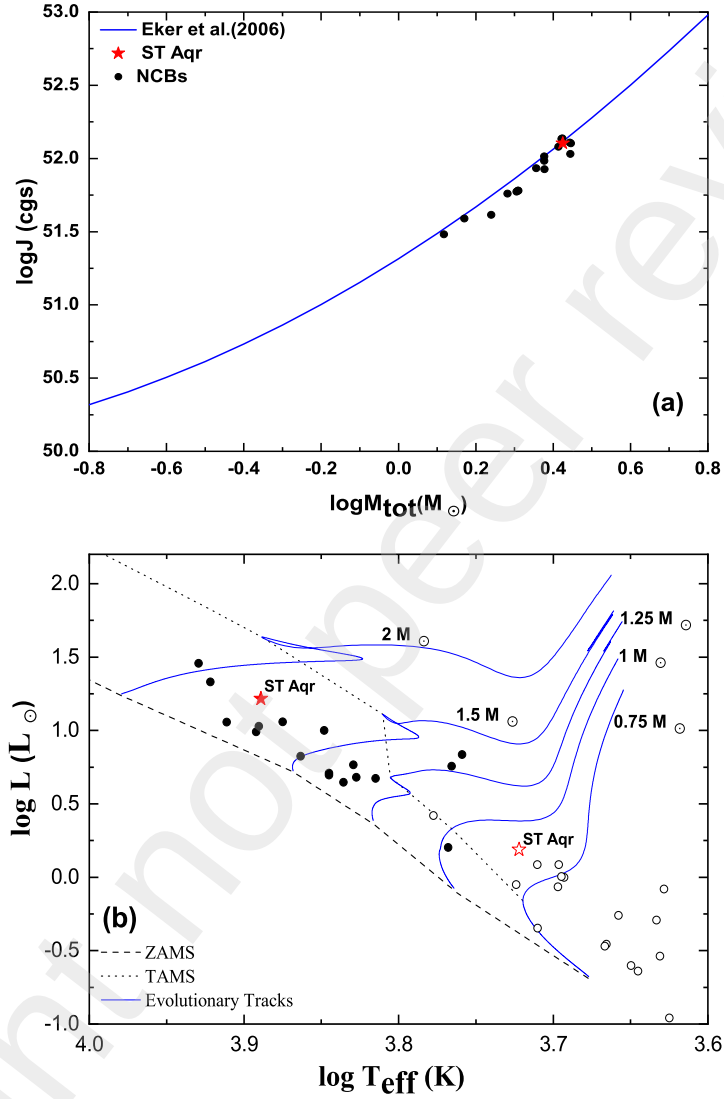


Figure 4: ST Aqr and the positions of NCBs in Table 5 on the $\log M_{\text{tot}} - \log J$ and HR diagrams. On the HR diagram, filled symbols represent the primary bodies, while hollow symbols represent the secondary bodies; the circle symbols indicate the NCBs taken from Table 5.

literature by elucidating the relationship between mass transfer processes and AML in the system. The combined analysis of future ground-based photometric and high-resolution spectroscopic observations of ST Aqr with data from TESS's subsequent observation sectors; will enable a more precise determination of ST Aqr's OP variations, the definitive identification of the spectral types, surface temperatures, and chemical abundances of its bodies, the confirmation of its evolutionary position on the H-R diagram, and a more reliable determination of effects such as AML, thereby providing significant insights into the physical and evolutionary structure of ST Aqr. Finally, observations must continue to track the future evolution of ST Aqr for the TRO test.

6. CRediT authorship contribution statement

M.F. Yıldırım: Methodology, Investigation, Resources, Analysis, Writing-original draft.

7. Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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